



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 8

TASKS :

1. Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.



```
#include <stdio.h>
#include <ctype.h>

void getUserInitials(char name[])
{
    printf("\nInitials: \"%c%c\"\n ", toupper(name[0]), toupper(name[6]));
}

int main()
{
    char name[] = "Virat Kohli";

    getUserInitials(name);

    return 0;
}
```

OUTPUT:

```
Initials: "VK"
```

2. Create a function `addToCart` that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function

Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.



```
#include <stdio.h>
#include <string.h>

void addToCart(char cart[][20], int *count, char product[])
{
    strcpy(cart[*count], product);
    (*count)++;

    printf("\nUpdated Cart:\n");

    for(int i = 0; i < *count; i++)
    {
        printf("%s\n", cart[i]);
    }
}

int main()
{
    char cart[10][20] = {"Laptop", "Mouse"};
    int count = 2;

    addToCart(cart, &count, "Keyboard");

    return 0;
}
```

OUTPUT:

```
Updated Cart:
Laptop
Mouse
Keyboard
```

3. Write two functions: `increaseFollowersByValue` and `increaseFollowersByReference`. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.



```
#include <stdio.h>

void increaseFollowersByValue(int followers)
{
    followers += 1000;
    printf("Inside Pass by Value = %d\n", followers);
}

void increaseFollowersByReference(int *followers)
{
    *followers += 1000;
    printf("Inside Pass by Reference = %d\n", *followers);
}

int main()
{
    int followers = 5000;

    increaseFollowersByValue(followers);
    printf("After Pass by Value = %d\n\n", followers);

    increaseFollowersByReference(&followers);
    printf("After Pass by Reference = %d\n", followers);

    return 0;
}
```

OUTPUT:

```
Inside Pass by Value = 6000
After Pass by Value = 5000

Inside Pass by Reference = 6000
After Pass by Reference = 6000
```

4. Build a function `formatPrice` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.



```
#include <stdio.h>

void formatPrice(int price)
{
    printf("₹%d\n", price);
}

int main()
{
    formatPrice(1599);
    formatPrice(24999);
    formatPrice(799);

    return 0;
}
```

OUTPUT:

```
₹1599
₹24999
₹799
```

5. Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).
Constraint: The refactored function should work for any string input, not just a specific use case.



```
#include <stdio.h>
#include <ctype.h>

void capitalizeFirstLetter(char name[])
{
    name[0] = toupper(name[0]);
}

int main()
{
    char product[] = "laptop";
    char username[] = "haaris";

    capitalizeFirstLetter(product);
    capitalizeFirstLetter(username);
}
```

```
printf("Product : %s\n",product);  
printf("Username : %s\n ",username);  
  
return 0;  
}
```

OUTPUT:

```
Product : Laptop  
Username : Haaris
```